Suraj Yadav

Software Engineer (Backend, DevOps & Security) | Distributed Systems | Cloud Native
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Education

Indian Institute of Technology Roorkee (IIT Roorkee) Sep 2020 - May 2024
Bachelor of Technology Roorkee, Uttarakhand

Technical Skills

Backend: Golang (Gin), Django Rest Framework, PostgreSQL, MongoDB, Postman, Redis, Kafka, REST APIs
Cloud & DevOps: AWS, Docker, Kubernetes, Git, GitHub Actions, Jenkins, CI/CD, ElasticSearch, Kibana, SigNoz, Trivy
Security & Systems: Threat Modeling, Secure SDLC, Malware Static/Dynamic Analysis, Debugging
Programming & Problem Solving: Golang(advanced; 250+ DSA problems solved), Python, C++ (Basic), Database Design

Experience

Mydbops **Nov 2025 - April 2026**
Associate Software Developer Remote

- Engineered resource-profiling check agents in Python/Go to automate database performance diagnostics.
- Integrated AWS CloudWatch and PMM APIs to track real-time RDS storage growth and alert on capacity exhaustion.
- Resolved critical engine bugs, implementing INNODB_SYS_TABLESPACES fallbacks for legacy Percona compatibility.
- Hardened secure API pipelines by implementing SSL/TLS exception handling for HashiCorp Vault credential fetches.
- Developed automated CLI tools to programmatically simulate database load spikes, table locks, and memory stress.

Globals ITeS **May 2025 - June 2025**
Security Research Intern Bangalore

- Reproduced CVE-2024-21413 in authorized lab to validate client-reported vulnerability and confirm scope.
- Developed defensive proof-of-concept in isolated environments to assess execution risk and recommend mitigations.
- Explored alternative delivery vectors and emulated firmware in the lab to analyse security gaps and improve detection.

Cyble Inc **June 2024 - Sep 2024**
Security Research Intern Bangalore

- Analyzed 50+ malware and phishing samples daily from VirusTotal and triage, identifying large-scale attack campaigns.
- Published research blogs on campaigns like OpenAI Sora and Zoom phishing, documenting techniques used by TAs.
- Authored 40+ advisories and weekly reports on CVEs, brute-force activity, and emerging threats for BFSI clients.
- Automated cloud-hosted T-POTs with SSH health checks, ensuring consistent log ingestion into Kibana dashboards.
- Built Python scripts to process 50K+ daily domains, filtering and flagging 100-200 high-risk entries for further analysis.

Cloud Security Club **May 2024 - June 2024**
Cloud Security Research Intern Remote

- Authored AWS security walkthrough for *flaws*, *CloudGoat* challenges, validated exploits and mitigations in isolated labs.
- Completed hands-on EC2, S3, IAM, Lambda audits in lab challenges; documented remediation guidance for learners.

Projects

logTraffic-Monitor | *github↗* **Sep 2025**

- Engineered a scalable log ingestion backend with Django, Kafka, and PostgreSQL, processing 10,000+ events/minute.
- Implemented Redis-based rate limiting at 100 requests/minute per API key, ensuring secure, multi-tenant API access.
- Integrated OpenTelemetry for distributed tracing and metrics, achieving 95% trace coverage across backend services.
- Automated log, trace, and metric exports to SigNoz, reducing incident detection time by 60% for cloud deployments.
- Optimized Kafka consumer throughput and reduced log persistence latency by 40% using asynchronous processing.
- Designed SigNoz dashboards and alerts, enabling proactive detection of errors, bottlenecks, API abuse in real time.

Secure File Sharing | *github↗* **Aug 2025**

- Engineered a backend in Golang, enabling Google OAuth and JWT authentication for 100% user session integrity.
- Designed PostgreSQL schemas and RBAC logic to enforce file access, reducing unauthorized access incidents by 95%
- Integrated AWS S3 with KMS encryption, ensuring all files are encrypted at rest, compliant with security best practices.
- Optimized file upload/download APIs, achieving 40% faster responses and seamless scalability for concurrent users.
- Enabled real-time file sharing and access management with resilient database setup, streamlining collaboration.

Restaurant Management | *github↗* **Aug 2025**

- Engineered RESTful APIs in Golang, scaling to 1k+ concurrent users and reducing median response latency by 35%.
- Designed MongoDB schemas for menus, orders, and invoices, optimizing queries to support 10k+ daily transactions.
- Automated CI/CD with GitHub Actions, ensuring 100% linting, security scans(Trivy, CodeQL), Docker image publishing.

- Implemented JWT-based authentication middleware, handling 1.5k+ requests/sec with secure, stateless authorization.
- Containerized backend and DB with Docker Compose, enabling zero-downtime deployment and faster cloud migration.
- Integrated health checks and monitoring, cutting incident response time by 30% and improving uptime to 99.99%.

Achievements / Extracurriculars

- Solved 250+ data structures and algorithms problems in Golang, strengthening backend problem-solving foundations.
- Core member of InfoSecIITR; ranked 2nd in India on CTFtime, won CSAW India Finals, 3rd JadeCTF 2022.
- Author of blogs documenting practical tutorials and challenge writeups: Docker, Jenkins, binary analysis, and forensics.
- Achieved 99.5%ile in JEE Mains 2020 among 1M+ students, AIR 9876 in JEE Advanced.